

Team SAMADHANA

Ikwath - Automated Kwatha (kadha) Preparation

Hand Calculations

Mechanical design to Biochemical outcome

Basis used in every step

Parameter	Value
Charge Volume	400 mL
Dose Volume	100 mL
Operating Temperature	88°C
Chamber Bore	100 mm
Pod Charge	25 g
Pod Ratio	1:16
Ambient Temperature	30°C
Ambient Humidity	60% RH
Water Heat Capacity (Cp)	4186 J/kg·K
SS304 Density / Cp	8000 kg/m ³ / 500 J/kg·K
Aluminium Density / Cp	2700 kg/m ³ / 900 J/kg·K

Chamber Volume Calculation

Given:

D = 100 mm
h = 127 mm

Find area

$$A = \frac{\pi D^2}{4}$$
$$A = \frac{3.1416(100)^2}{4}$$
$$A = 7854 \text{ mm}^2$$

Find volume

$$V = A \times h$$
$$V = 7854 \times 127$$
$$V = 997,458 \text{ mm}^3$$

Since,

$$1 \text{ mL} = 1000 \text{ mm}^3$$

$$V \approx 997.5 \text{ mL} \approx 1 \text{ L}$$

Result: Chamber capacity $\approx 1 \text{ L}$.

Liquid Depth at Each Stage

Given:

$$D = 100 \text{ mm}$$
$$A = \frac{\pi D^2}{4} = \frac{3.1416(100)^2}{4} = 7854 \text{ mm}^2$$

Since,

$$V = A \times h$$

Therefore,

$$h = \frac{V}{A}$$

Volume	Calculation	Liquid Depth
1000 mL	1000000/7854	127.3 mm
400 mL	400000/7854	50.9 mm
300 mL	300000/7854	38.2 mm
100 mL	100000/7854	12.7 mm

Result:

400 mL → **50.9 mm** depth

300 mL → **38.2 mm** depth

100 mL → **12.7 mm** depth

Chamber Weight

Given:

- Inner SS = 1.5 mm
- Aluminium = 3 mm
- Outer SS = 1.5 mm
- Wall height = 150 mm
- Base = 6 mm
- SS density = $8.0 \times 10^{-3} \text{ g/mm}^3$
- Al density = $2.7 \times 10^{-3} \text{ g/mm}^3$

1. Inner SS

$$V = \pi(51.5^2 - 50^2)(150)$$

$$V \approx 71,745 \text{ mm}^3$$

$$m = 71,745(8 \times 10^{-3})$$

$$m \approx 574 \text{ g}$$

2. Aluminium core

$$V = \pi(54.5^2 - 51.5^2)(150)$$

$$V \approx 149,850 \text{ mm}^3$$

$$m = 149,850(2.7 \times 10^{-3})$$

$$m \approx 405 \text{ g}$$

3. Outer SS

$$V = \pi(56^2 - 54.5^2)(150)$$

$$V \approx 78,108 \text{ mm}^3$$

$$m = 78,108(8 \times 10^{-3})$$

$$m \approx 625 \text{ g}$$

4. Base

$$m_{\text{base}} \approx 316 \text{ g}$$

Total Chamber Weight

$$m_{\text{total}} = 574 + 405 + 625 + 316$$

$$m_{\text{total}} \approx 1920 \text{ g} = 1.92 \text{ kg}$$

Result: Chamber weight $\approx 1.92 \text{ kg}$

Heat Capacity of Chamber

$$C_{\text{cham}} = m_{\text{SS}}C_{p,\text{SS}} + m_{\text{Al}}C_{p,\text{Al}}$$

Using:

- $m_{\text{SS}} = 1.435 \text{ kg}$
- $C_{p,\text{SS}} = 500 \text{ J/kgK}$
- $m_{\text{Al}} = 0.484 \text{ kg}$
- $C_{p,\text{Al}} = 900 \text{ J/kgK}$

$$C_{\text{cham}} = (1.435)(500) + (0.484)(900) \\ = 717.5 + 435.6$$

$$C_{\text{cham}} \approx 1154 \text{ J/K}$$

Result: Chamber itself requires approximately **1154 J of heat for every 1°C temperature rise.**

Lid & Chamber Fit

Given:

Skirt ID = 113 mm

Rolled rim OD = 114 mm

$$\text{Clearance} = \text{ID} - \text{OD}$$

$$= 113 - 114$$

$$= -1.0 \text{ mm}$$

Result: No Interference fit

Required approximately:

- Skirt ID = **115 mm**
- Lid OD = **118 mm**

Heating Energy & Time

Energy Required

1. Heat required for water

$$\begin{aligned}Q_{\text{water}} &= mC_p\Delta T \\&= 0.400(4186)(58) \\Q_{\text{water}} &= 97.1 \text{ kJ}\end{aligned}$$

2. Heat required for chamber

$$\begin{aligned}Q_{\text{chamber}} &= C_{\text{cham}}\Delta T \\&= 1154(58) \\Q_{\text{chamber}} &= 66.9 \text{ kJ}\end{aligned}$$

3. Energy for evaporation

$$\begin{aligned}Q_{\text{evap}} &= mh_{\text{fg}} \\&= 0.300(2288.5) \\Q_{\text{evap}} &= 686.6 \text{ kJ}\end{aligned}$$

Total Energy

$$\begin{aligned}Q_{\text{total}} &= 97.1 + 66.9 + 686.6 \\Q_{\text{total}} &= 850.6 \text{ kJ}\end{aligned}$$

Result: About 81% of the energy goes into evaporation for 1 cycle.

Heating Time

Using heater power:

$$\begin{aligned}P &= 850 \text{ W} \\t &= \frac{Q}{P} \\&= \frac{850600}{850}\end{aligned}$$

$$t \approx 1001 \text{ s}$$

$$t \approx 16.7 \text{ min}$$

Important: The original text's 193 s / 3.2 min is not consistent with 850.6 kJ at 850 W. The correct ideal energy-only time is about 16.7 min.

Heat Loss to Room

$$\begin{aligned}h_{\text{conv}} &= 6.6 \text{ W/m}^2\text{K} \\h_{\text{rad}} &= 2.1 \text{ W/m}^2\text{K} \\A &= 0.0528 \text{ m}^2 \\\Delta T &= 55^\circ\text{C} \\Q_{\text{loss}} &= (h_{\text{conv}} + h_{\text{rad}})A\Delta T \\&= (6.6 + 2.1)(0.0528)(55) \\Q_{\text{loss}} &\approx 25.3 \text{ W}\end{aligned}$$

So heat loss is relatively small compared with the heater power.

Bottom Temperature

Given:

$$A = 78.5 \text{ cm}^2 = 0.00785 \text{ m}^2$$
$$h \approx 5000 \text{ W/m}^2\text{K}$$

At 600 W

$$q'' = \frac{600}{0.00785}$$
$$\boxed{q'' = 76.4 \text{ kW/m}^2}$$
$$\Delta T = \frac{q''}{h}$$
$$= \frac{76400}{5000}$$
$$\boxed{\Delta T = 15.3 \text{ K}}$$

At 800 W

$$\boxed{\Delta T = 20.4 \text{ K}}$$

At 1000 W

$$\boxed{\Delta T = 25.5 \text{ K}}$$

Therefore, at 1 kW:

$$T_{\text{bottom}} = 88 + 25.5$$
$$\boxed{T_{\text{bottom}} \approx 113^\circ\text{C}}$$

Powder Settling on Base

Given:

$$k_{\text{bed}} = 0.4 \text{ W/mK}$$
$$q'' = 101900 \text{ W/m}^2$$
$$L = 0.003 \text{ m}$$

Using:

$$\Delta T = \frac{q''L}{k}$$
$$= \frac{101900(0.003)}{0.4}$$
$$\boxed{\Delta T \approx 764 \text{ K}}$$

Result: 3 mm settled powder layer can cause **very high temperature rise**.

Keep the pod **at least 15 mm above the base**.

Why Reduction Is Slow

Does Water Boil at 88°C?

$$P_{\text{sat}@88^{\circ}\text{C}} = 65.0 \text{ kPa}$$

$$P_{\text{atm}} = 101.3 \text{ kPa}$$

Since:

$$65.0 < 101.3$$

$$\boxed{\text{Water does not boil at } 88^{\circ}\text{C}}$$

Therefore: Water removal occurs mainly through **evaporation/diffusion**, not boiling.

Vapour Leaving Liquid Surface

Calculated surface conductance:

$$\boxed{G_s = 0.116 \text{ L/s}}$$

Result: Surface can release vapour at approximately **0.116 L/s** under the assumed conditions.

Vapour Leaving Through Lid

Given:

- 12 × Ø4 mm holes
- 1 × Ø20 mm outlet
- Total opening area ≈ **465 mm²**
- Stack height = **50 mm**

Calculated:

$$v \approx 0.740 \text{ m/s}$$

$$G_v \approx 0.213 \text{ L/s}$$

$$\boxed{G_v = 0.213 \text{ L/s}}$$

Result: Vent conductance ≈ **0.213 L/s**.

So, in simple terms:

Liquid surface: 0.116 L/s

Lid vent: 0.213 L/s

Surface evaporation is the more restrictive step.

Actual Evaporation Rate

Surface and vent act like two resistances:

$$G_{\text{total}} = \left(\frac{1}{G_s} + \frac{1}{G_v} \right)^{-1}$$

Given:

$$G_s = 0.116 \text{ L/s}$$

$$G_v = 0.213 \text{ L/s}$$

$$G_{\text{total}} = \left(\frac{1}{0.116} + \frac{1}{0.213} \right)^{-1}$$

$$\boxed{G_{\text{total}} \approx 0.075 \text{ L/s}}$$

Vapour density difference:

$$\begin{aligned}\Delta\rho &= 0.390 - 0.018 \\ &= 0.372 \text{ kg/m}^3\end{aligned}$$

Therefore,

$$\begin{aligned}\dot{m} &= G_{\text{total}}\Delta\rho \\ &= 0.075(0.372) \\ \dot{m} &\approx 0.0280 \text{ g/s}\end{aligned}$$

For 275 g water:

$$\begin{aligned}t &= \frac{275}{0.0280} \\ t &= 9821 \text{ s} \\ t &\approx 164 \text{ min}\end{aligned}$$

Result: Reduction takes approximately 164 min.

Is the Heater the Limitation?

Heater power:

$$\dot{Q} = 800 \text{ W}$$

Maximum evaporation rate:

$$\begin{aligned}\dot{m}_{\text{max}} &= \frac{\dot{Q}}{h_{\text{fg}}} \\ &= \frac{800}{2,288,500} \\ &= 0.000350 \text{ kg/s} \\ \dot{m}_{\text{max}} &= 0.350 \text{ g/s}\end{aligned}$$

Actual transport rate:

$$\dot{m} = 0.028 \text{ g/s}$$

Ratio:

$$\frac{0.350}{0.028} \approx 12.5$$

Result: Heater capacity is about 12.5× higher than the transport capacity.

Will a Fan Solve It?

If vent resistance becomes negligible:

$$\begin{aligned}\dot{m}_{\text{max}} &= G_s(\rho_{v,s} - \rho_{v,\infty}) \\ &= 0.116(0.390 - 0.018) \\ \dot{m} &\approx 0.0432 \text{ g/s}\end{aligned}$$

Time:

$$\begin{aligned}t &= \frac{275}{0.0432} \\ t &\approx 106 \text{ min}\end{aligned}$$

Result: Even with unlimited venting, reduction takes about 106 min.

What Surface Area Is Needed?

Target:

$$\dot{m} = 0.25 \text{ g/s}$$

Required area:

$$A_{\text{req}} = \frac{\dot{m}}{k_m(\rho_{v,s} - \rho_{v,\infty})}$$

Calculated:

$$A_{\text{req}} \approx 454 \text{ cm}^2$$

Equivalent circular diameter:

$$A = \frac{\pi D^2}{4}$$
$$D = \sqrt{\frac{4A}{\pi}}$$
$$D \approx 241 \text{ mm}$$

Result: A 241 mm diameter pan would be required for a 100 mL dose to reduce in about 20 min.

Pressure Reduction

Pressure Required for Boiling at 88°C

Approximate saturation relationship:

Pressure Boiling Point

55 kPa 83.7°C

60 kPa 85.9°C

65 kPa 88.0°C

70 kPa 89.9°C

75 kPa 91.8°C

Therefore:

$$P \approx 65 \text{ kPa}$$

At 65 kPa absolute pressure, water boils at approximately 88°C.

Reduction Time When Boiling

Now evaporation is controlled by heater power.

$$\begin{aligned} t &= \frac{m h_{fg}}{\dot{Q}} \\ &= \frac{0.275(2,288,500)}{800} \\ &= 787 \text{ s} \\ t &\approx 13.1 \text{ min} \end{aligned}$$

Result:

164 min → 13.1 min

Approximately:

$$12.5 \times \text{faster}$$

while maintaining approximately 88°C.

Pressure Control

Given:

$$\frac{dT_{\text{sat}}}{dP} \approx 0.40 \text{ K/kPa}$$

For:

$$T = 88 \pm 1.5^\circ\text{C}$$

Required pressure variation:

$$\Delta P = \frac{1.5}{0.40}$$

$$\Delta P \approx 3.75 \text{ kPa}$$

So approximately:

$$P = 65 \pm 3.7 \text{ kPa}$$

Result: Controlling pressure around 65 kPa absolute can maintain the boiling temperature near 88°C.

Vacuum System Requirement

Vapour generation

$$\dot{m} = \frac{800}{2,288,500}$$
$$\dot{m} = 0.350 \text{ g/s}$$

Approximately:

$$V_{\text{vapour}} \approx 0.90 \text{ L/s}$$

Condenser

For 800 W heat removal:

$$A = \frac{800}{50 \times 40}$$

$$A \approx 0.40 \text{ m}^2$$

Vacuum pull-down

Required volume:

$$V \approx 0.28 \text{ L}$$

At pump flow:

$$2 \text{ L/min}$$

$$t \approx 8.4 \text{ s}$$

Result: A small diaphragm vacuum pump (~10 W) with an air-cooled condenser can theoretically handle the required duty.

Main Engineering Conclusion

At 88°C and atmospheric pressure → reduction is very slow

At 65 kPa absolute → water boils at 88°C

Reduction time: 164 min → 13.1 min

So the key design change is pressure reduction, not simply increasing heater power.

Vacuum Line & Weighing System

Vacuum pressure difference creates a force on the weighing system.

$$F = \Delta P \times A_{\text{tube}}$$

Apparent mass error:

$$m_{\text{error}} = \frac{F}{g}$$

Calculated error

Vacuum Line Bore	Apparent Mass Error	After ±1 kPa Compensation
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Ø3 mm	26 g	0.7 g
Ø6 mm	105 g	2.9 g
Ø20 mm	1163 g	32 g

Result: Use a **small Ø3 mm vacuum line** and weigh the **complete vacuum envelope**.

Herb Extraction (Biochemical calculations)

How Fast Does Extraction Happen?

Diffusion coefficient at 88°C:

$$D_{88} = D_{25} \left(\frac{T_{88}}{T_{25}} \right) \left(\frac{\mu_{25}}{\mu_{88}} \right)$$

Using the given values:

$$D_{88} = 1.0 \times 10^{-9} (1.211) (2.730)$$
$$D_{88} = 3.31 \times 10^{-9} \text{ m}^2/\text{s}$$

Effective diffusion coefficient:

$$D_{\text{eff}} = D_{88} \frac{\epsilon}{\tau}$$
$$= 3.31 \times 10^{-9} \frac{0.5}{3.0}$$
$$D_{\text{eff}} = 5.51 \times 10^{-10} \text{ m}^2/\text{s}$$

Extraction time

Particle Size	95% Extraction	99% Extraction
0.5 mm	0.5 min	0.8 min
1.0 mm	1.9 min	3.2 min
1.5 mm	4.3 min	7.1 min
2.0 mm	7.7 min	12.6 min
4.0 mm	30.6 min	50.3 min

Result: For typical **1.5–2 mm particles**, ~99% extraction takes about **7–13 min**.

The long cycle time is mainly due to **water reduction/evaporation**, not extraction.

Does Stirring Improve Extraction?

Given:

$$\text{Re} = 445$$
$$\text{Sc} = 102$$

Sherwood number:

$$\text{Sh} = 2 + 0.6 \text{Re}^{1/2} \text{Sc}^{1/3}$$
$$\text{Sh} \approx 61.1$$

Mass-transfer coefficient:

$$k_f \approx 1.35 \times 10^{-4} \text{ m/s}$$

Mass-transfer Biot number:

$$\text{Bi}_m = \frac{k_f R}{D_{\text{eff}}}$$
$$\text{Bi}_m \approx 183$$

Since:

$$\text{Bi}_m \gg 1$$

Result: External mass transfer is not the main limitation.

Stirring does not significantly speed up extraction.

Its main purpose is to **keep the solids moving and prevent them from settling on the hot base**.

What Happens with a Sealed Pod?

For a sealed pod:

$$t_c = \frac{4L^2}{\pi^2 D_{\text{bed}}}$$

Given:

$$D_{\text{bed}} = 6.61 \times 10^{-10} \text{ m}^2/\text{s}$$

Effect of pod thickness

Pod Half-Thickness	95% Extraction Time
1 mm	28 min
2 mm	114 min
3 mm	256 min
5 mm	711 min

Result: A thick sealed pod creates a very long diffusion path.

Thin / open-flow pod design is preferred.

Flow Required Through Pod Bed

Bed permeability:

$$k = \frac{\epsilon^3 d^2}{180(1 - \epsilon)^2}$$

Calculated:

$$k = 2.22 \times 10^{-9} \text{ m}^2$$

Using Darcy's law:

$$u = \frac{k\Delta P}{\mu L}$$

For a **10 mm bed**:

Pressure Drop	Darcy Velocity	Bed Turnover
2 Pa	1.4 mm/s	7.3 s
5 Pa	3.4 mm/s	2.9 s
20 Pa	13.6 mm/s	0.7 s

Result: Only a few pascals pressure drop can circulate the bed within a few seconds.

A **small recirculation loop** is sufficient.

Extract Lost with Spent Pod

Given:

Pod charge:

$$W = 25 \text{ g}$$

Liquid retained by pod:

$$2.5 \text{ g/g}$$

Therefore,

$$R = 25(2.5)$$

$$R = 62.5 \text{ g}$$

Recovery comparison

Method	Recovery
No press	84.4%
300 g press + 100 g rinse	92.0%
60% press + rinse	93.8%
Press + rinse	98.3%

Result: Without pressing, about **16% of extractive is lost** with the spent pod.
Press + rinse significantly improves recovery.

Water Balance

Initial water:

400 g

Final dose:

100 g

Residual water:

25 g

Therefore, water to remove:

$$E = 400 - 100 - 25$$

$$E = 275 \text{ g}$$

Result: 275 g of water must be removed during reduction.

Does Faster Heating Damage the Chemistry?

Retention is calculated using:

$$\text{Retention} = e^{-k}$$

Comparison:

Activation Energy	Classical Cycle	Faster 88°C Cycle
50 kJ/mol	57.5%	87.7%
80 kJ/mol	64.1%	92.0%
110 kJ/mol	76.5%	96.1%

Result: Under these assumed degradation parameters, the **shorter 88°C cycle gives higher retention.**

Volatile Aromatic Loss

Volatile retention depends on the volatility parameter α .

α	Volatile retained
0.1	87%
0.3	66%
0.5	50%
1.0	25%
2.0	6%

Result: More volatile compounds are more likely to be lost during water reduction.
Vacuum boiling should therefore be evaluated for aroma retention experimentally.

Sensors & Control

Is the Temperature Sensor in the Liquid?

Given:

Chamber height:

$$150 \text{ mm}$$

Sensor immersion:

$$120 \text{ mm}$$

Distance from base:

$$150 - 120 = 30 \text{ mm}$$

Cross-sectional area:

$$A = 7854 \text{ mm}^2$$

Volume below sensor tip:

$$\begin{aligned} V &= A \times 30 \\ &= 7854(30) \\ \boxed{V &\approx 236 \text{ mL}} \end{aligned}$$

Result: Below approximately **236 mL**, the sensor tip becomes exposed to vapour.
Increase immersion length to approximately **144 mm**.

Temperature Overshoot

Thermal response time:

$$\tau \approx 5.5 \text{ s}$$

Temperature lag:

$$\Delta T_{\text{lag}} = \frac{C_{\text{tot}} \tau}{\text{thermal input}}$$

Given calculation:

$$\boxed{\Delta T_{\text{lag}} \approx 1.95 \text{ K}}$$

Stored heater energy contribution:

$$\begin{aligned} \Delta T_{\text{stored}} &= \frac{285(25)}{2828} \\ \boxed{\Delta T_{\text{stored}} &\approx 2.52 \text{ K}} \end{aligned}$$

Total:

$$\begin{aligned} \Delta T &\approx 1.95 + 2.52 \\ \boxed{\Delta T &\approx 4.5^\circ\text{C}} \end{aligned}$$

Result: Temperature can overshoot by roughly **4–4.5°C**.
Pressure-controlled boiling can reduce this problem because boiling temperature is set by pressure.

Can Weight Detect the Endpoint?

Assumed solids partial specific volume:

$$0.62 \text{ mL/g}$$

Calculated final mass for 100 mL:

Pod / Yield	Final Mass
10 g at 15%	100.3 g

25 g at 20%	101.6 g
25 g at 25%	102.1 g

Result: Solids slightly increase the final measured mass.
Gravimetric endpoint correction is required.

Load Cell Thermal Drift

Heat conducted to load cell:

$$Q_{\text{cond}} = \frac{kA\Delta T}{L}$$

Given calculation:

$$Q_{\text{cond}} \approx 3.26 \text{ W}$$

Including radiation:

$$Q_{\text{total}} \approx 4.3 \text{ W}$$

Temperature rise:

$$\Delta T \approx 15.9 \text{ K}$$

Load-cell drift:

$$= 0.0002(15.9)(5000)$$

$$\approx 16 \text{ g}$$

Result: Direct thermal exposure can cause about **16 g error**.

Use **PEEK standoffs + radiation shield** to reduce this to approximately **0.8 g**.

Load Cell Capacity

Estimated maximum load:

$$1.92 + 0.30 + 0.45 + 0.40$$

$$= 3.07 \text{ kg}$$

For a 5 kg load cell:

$$\text{Margin} = \frac{5}{3.07}$$

$$\text{Margin} \approx 1.63 \times$$

Result: A 5 kg load cell provides approximately **1.63× capacity margin** for the stated load assumptions.

Final Cycle

Recommended Cycle Time

Stage	Time	Cumulative
Fill + verify 400 mL	0.5 min	0.5 min
Heat to 88°C + pull to 65 kPa	2.5 min	3.0 min
Reduction — remove 275 g	13.1 min	16.1 min
Lift pod + press + drain	1.0 min	17.1 min
Gravimetric trim	1.0 min	18.1 min
Vent + cool + dispense	1.5 min	19.6 min

Total cycle $\approx$ 20 min

Design Options

Option	Cycle	Peak Temp
As drawn	170 min	88°C
Open pan	112 min	88°C
Fan + larger surface	48 min	88°C
Vacuum at 65 kPa	20 min	88°C
Atmospheric boiling	19 min	100°C
Classical reference	60–90 min	97°C

Key result:

65 kPa vacuum + 88°C $\rightarrow$ 20 min cycle

Drawing Changes, we undergone and iterated:

Part	Feature	Current	Revised
AKPD-01	Lid skirt ID	Ø113	Ø115 mm
AKPD-01	Lid OD	Ø116	Ø118 mm
AKPD-01	Vent area	465 mm ²	1400 mm² / fan
AKPD-02	Demister	Present	Delete
AKPD-03	Immersion	120 mm	144 mm
AKPD-03	Sensor sheath	Ø6 mm	Ø3 mm
AKPD-04	Stir bar	50 × Ø10	35 × Ø8 mm
AKPD-05	Wall	6 mm	2.5 mm
AKPD-06	Plate	Ø118	Ø140 mm
AKPD-06	Standoff	SS304	PEEK / ceramic